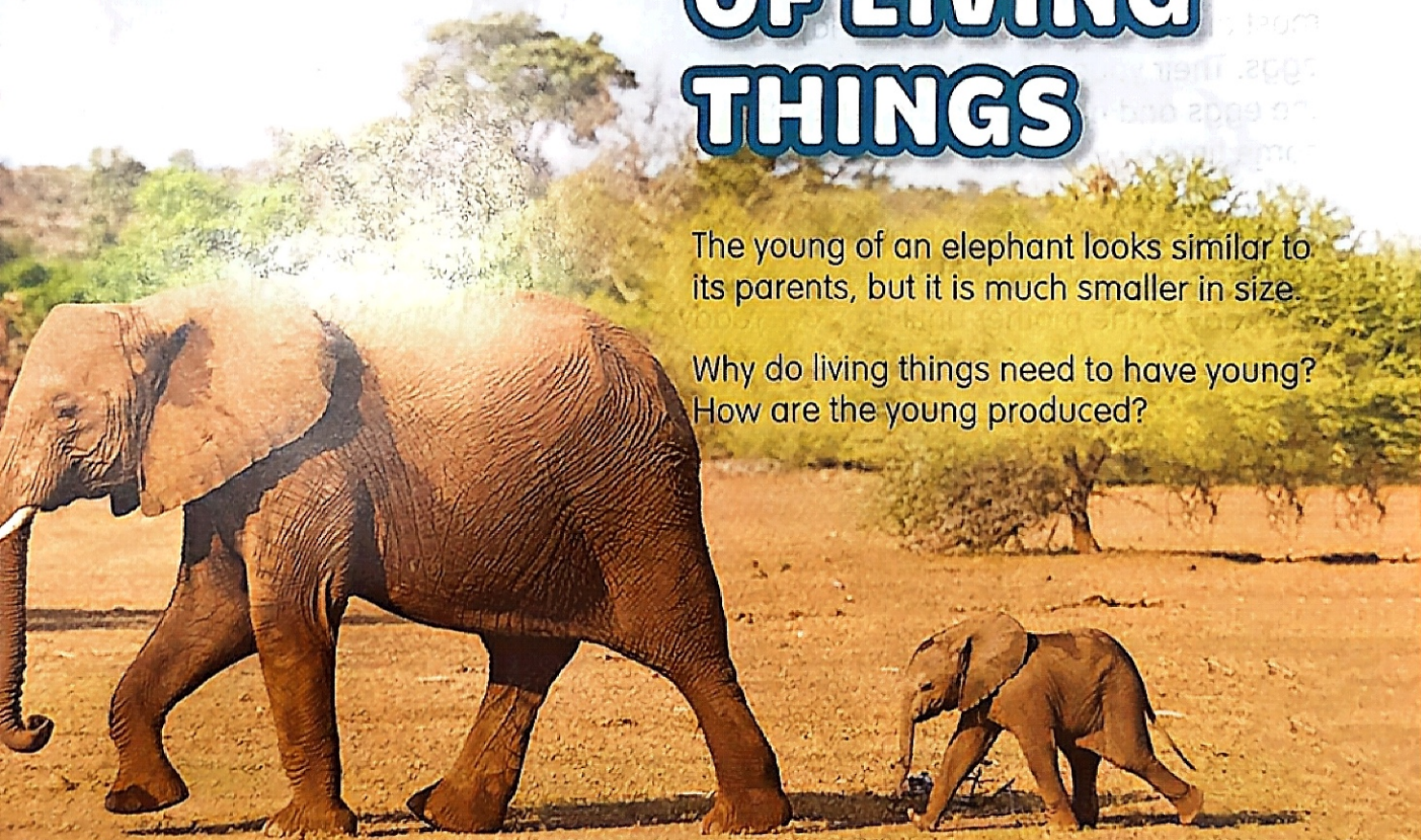


UNIT 3

LIFE CYCLES OF LIVING THINGS

The young of an elephant looks similar to its parents, but it is much smaller in size.

Why do living things need to have young?
How are the young produced?



Let's Find Out:

- How do living things reproduce?
- What is a life cycle?
- What are the life cycles of some living things?
- Why is it important to study the life cycles of living things?

3.1

How Do Living Things Reproduce?

Living things **reproduce** and have young.

Different living things reproduce in different ways. For example, some animals lay eggs, while some animals give birth to young alive.

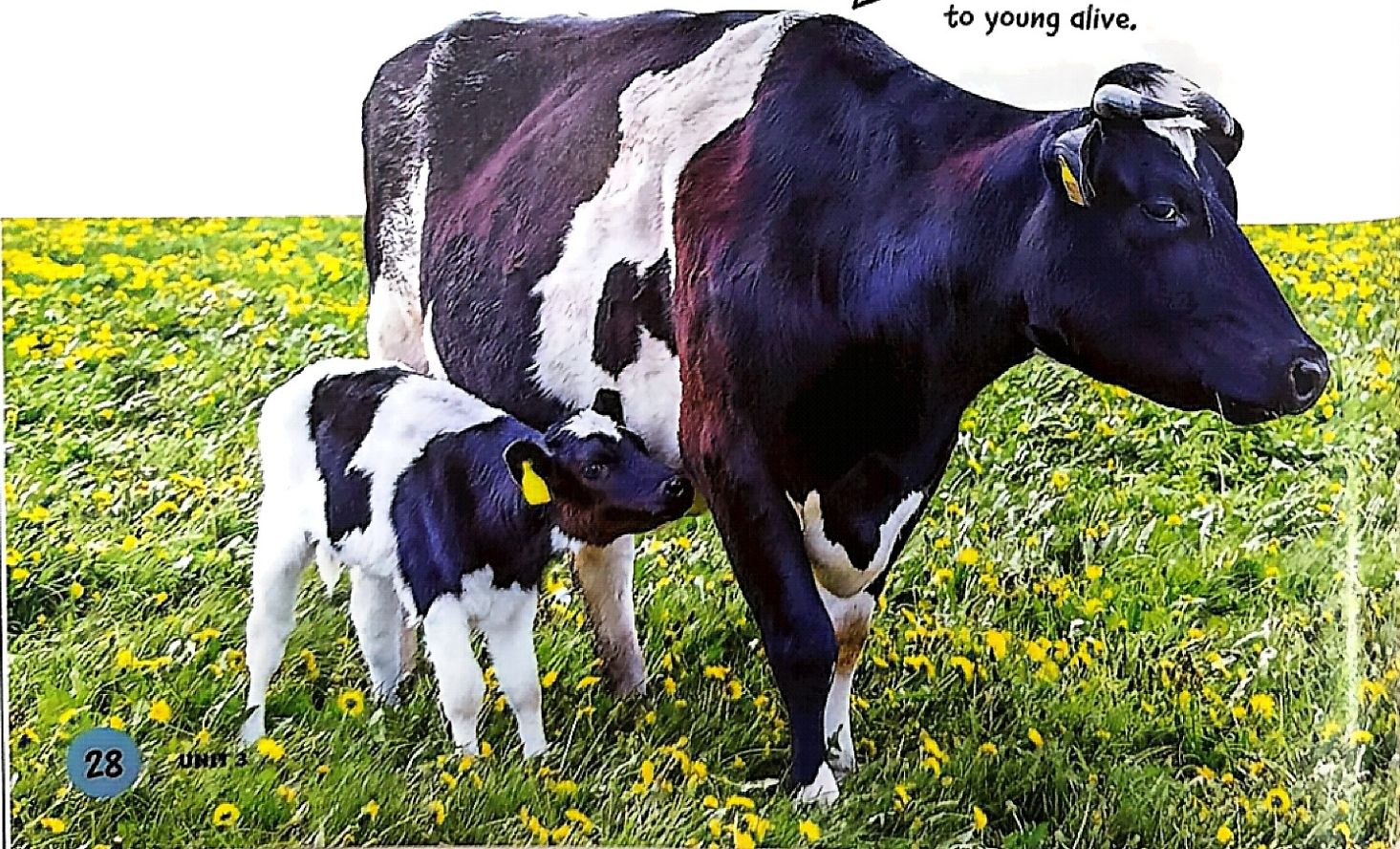
Fish, birds, insects, most reptiles and most amphibians reproduce by laying eggs. Their young **develop** inside the eggs and hatch from the eggs after some time.

Most mammals reproduce by giving birth to young alive. Their young develop inside the body of the mother until they are ready to be born.



Some birds lay their eggs in a nest.

Female cows give birth to young alive.

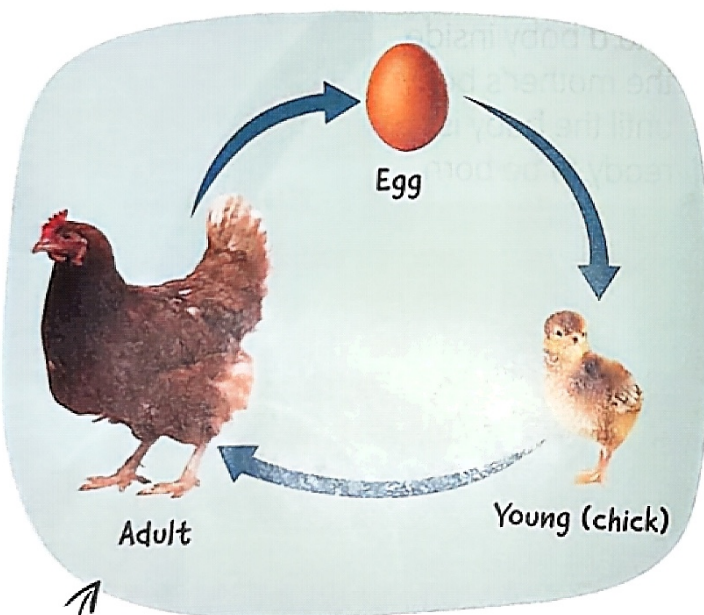


What is a Life Cycle?

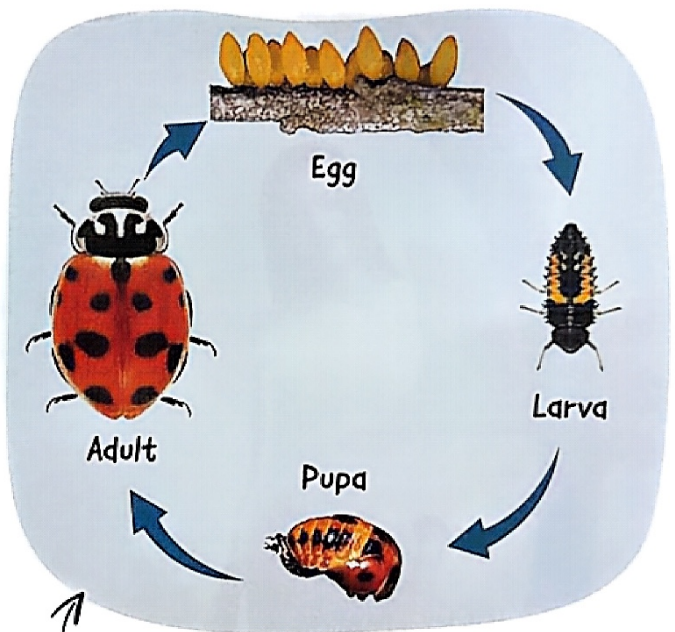
All living things go through a cycle called a **life cycle**.

The young of a living thing goes through a similar life cycle as its parents. In this way, living things of the same kind continue to be found on the Earth.

A life cycle is made up of different **stages** of growth and development. The young of some living things look similar to the adult, while the young of other living things do not.



A chicken has three stages in its life cycle. The chick looks like the adult, but it is smaller.



A ladybird beetle has four stages in its life cycle. The larva and pupa look very different from the adult.



QUICK CHECK

1. Why do living things reproduce?
2. State two ways by which animals produce their young.
3. What is a life cycle?

3.3

What Are the Life Cycles of Some Living Things?

Life cycle of a human

The life cycle of a human is made up of three stages — egg, young and adult.

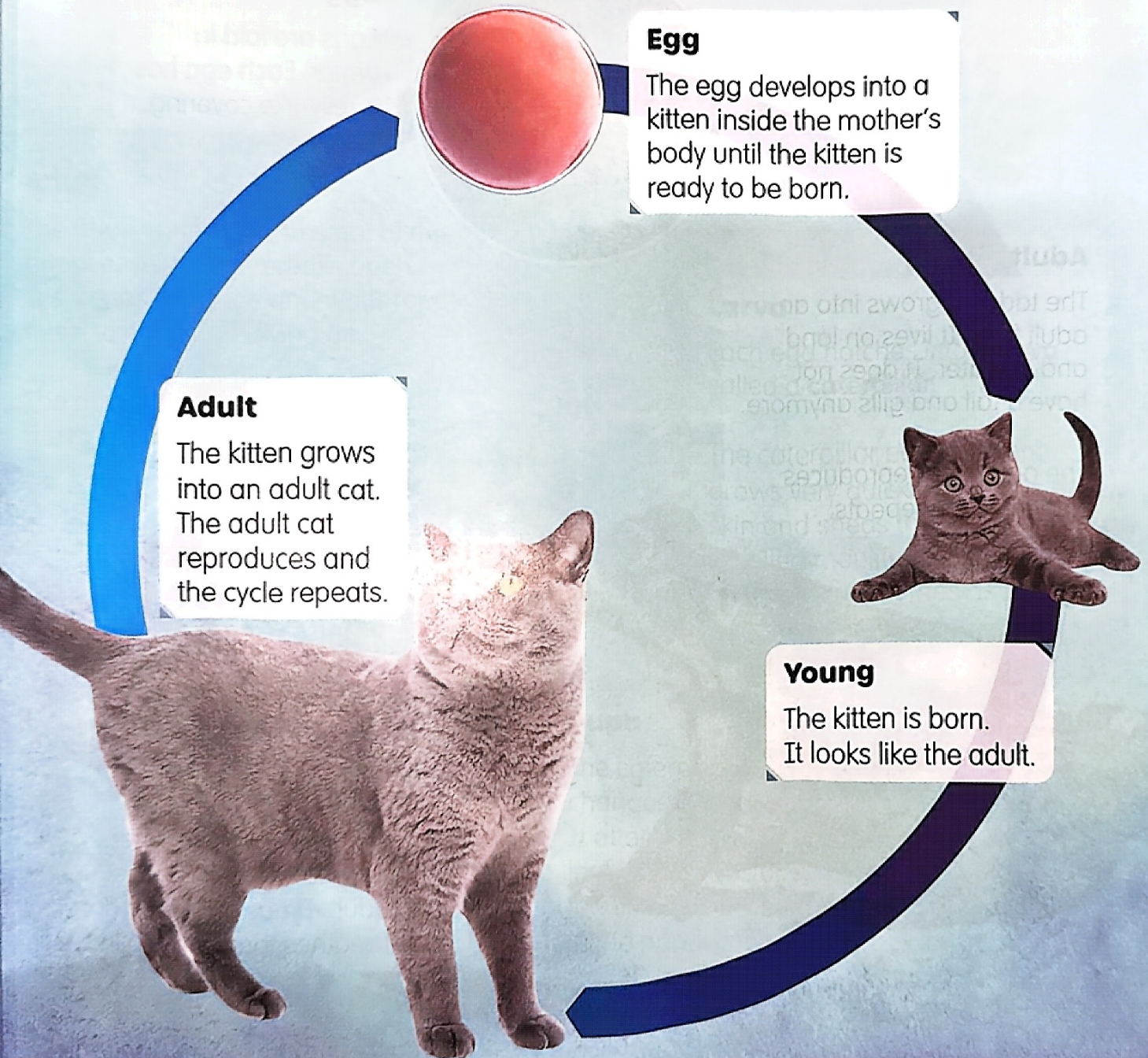


Activity Book

Activity 1, pages 25-26

Life cycle of a cat

The life cycle of a cat is made up of three stages — egg, young and adult.



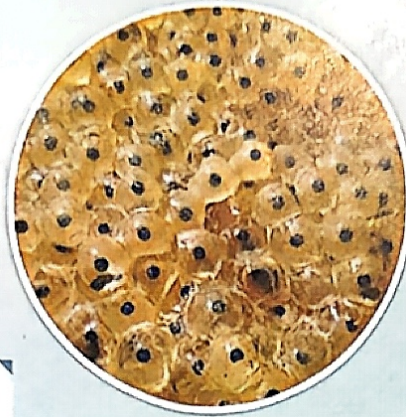
Life cycle of a cat

Life cycle of a frog

The life cycle of a frog is made up of three stages — egg, young and adult.

Egg

Eggs are laid in water. Each egg has a jelly-like covering.



Adult

The tadpole grows into an adult frog. It lives on land and in water. It does not have a tail and gills anymore.

The adult frog reproduces and the cycle repeats.



Young

Each egg hatches into a **tadpole**.

A tadpole lives in water. It does not look like the adult. It has a tail and breathes through its gills.



Life cycle of a frog

Life cycle of a butterfly

The life cycle of a butterfly is made up of four stages — egg, larva, pupa and adult.

Egg

Eggs are laid on plants.



Larva

Each egg hatches into a **larva** called a **caterpillar**.

The caterpillar eats a lot and grows very quickly. It grows a new skin and sheds the old one. This is called moulting. The caterpillar **moults** several times as it grows.

Pupa

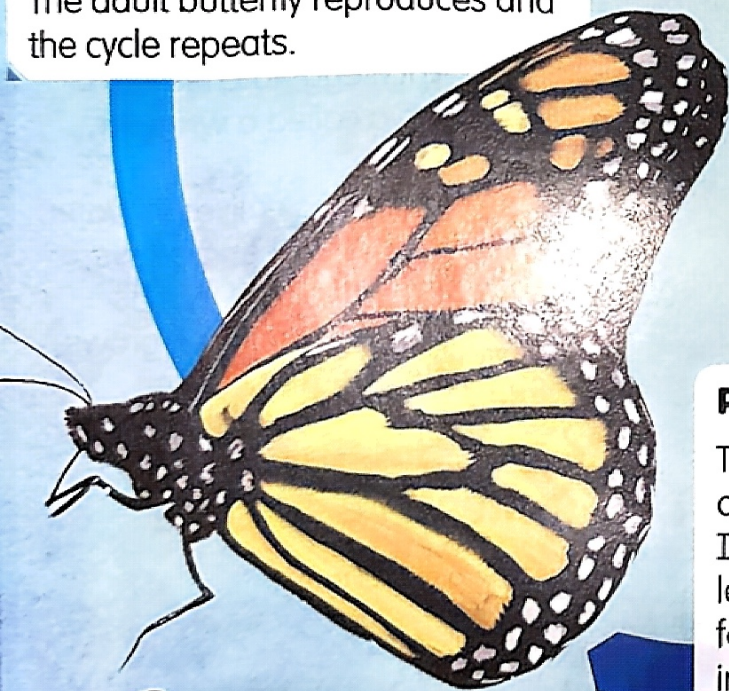
The caterpillar changes into a **pupa**. It attaches itself to a leaf or twig. It stops feeding and develops in the pupal case.



Adult

The adult butterfly comes out of the pupal case when it breaks open. It spreads its wings and waits for them to dry before flying off.

The adult butterfly reproduces and the cycle repeats.



RESEARCH

Silk comes from the outer covering of the silk moth's pupa. Find out how silk is produced.

Life cycle of a butterfly



Activity Book

Activity 2, pages 27-28

Life cycle of a mosquito

The life cycle of a mosquito is made up of four stages — egg, larva, pupa and adult.

Adult

The adult mosquito comes out of the pupal case when it breaks open.

The adult mosquito reproduces and the cycle repeats.

Egg

Eggs are laid in still water.



Larva

Each egg hatches into a larva called a **wiggler**.

The wiggler lives in water. It eats a lot and grows very quickly. It moults several times as it grows.



Pupa

The wiggler changes into a pupa. It lives near the water surface. It stops feeding and develops in the pupal case.

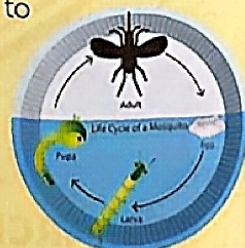


Life cycle of a mosquito



CREATIVE SCIENCE

Make a model to show the life cycle of a mosquito.



Activity Book

► Creative Science, pages 29-30

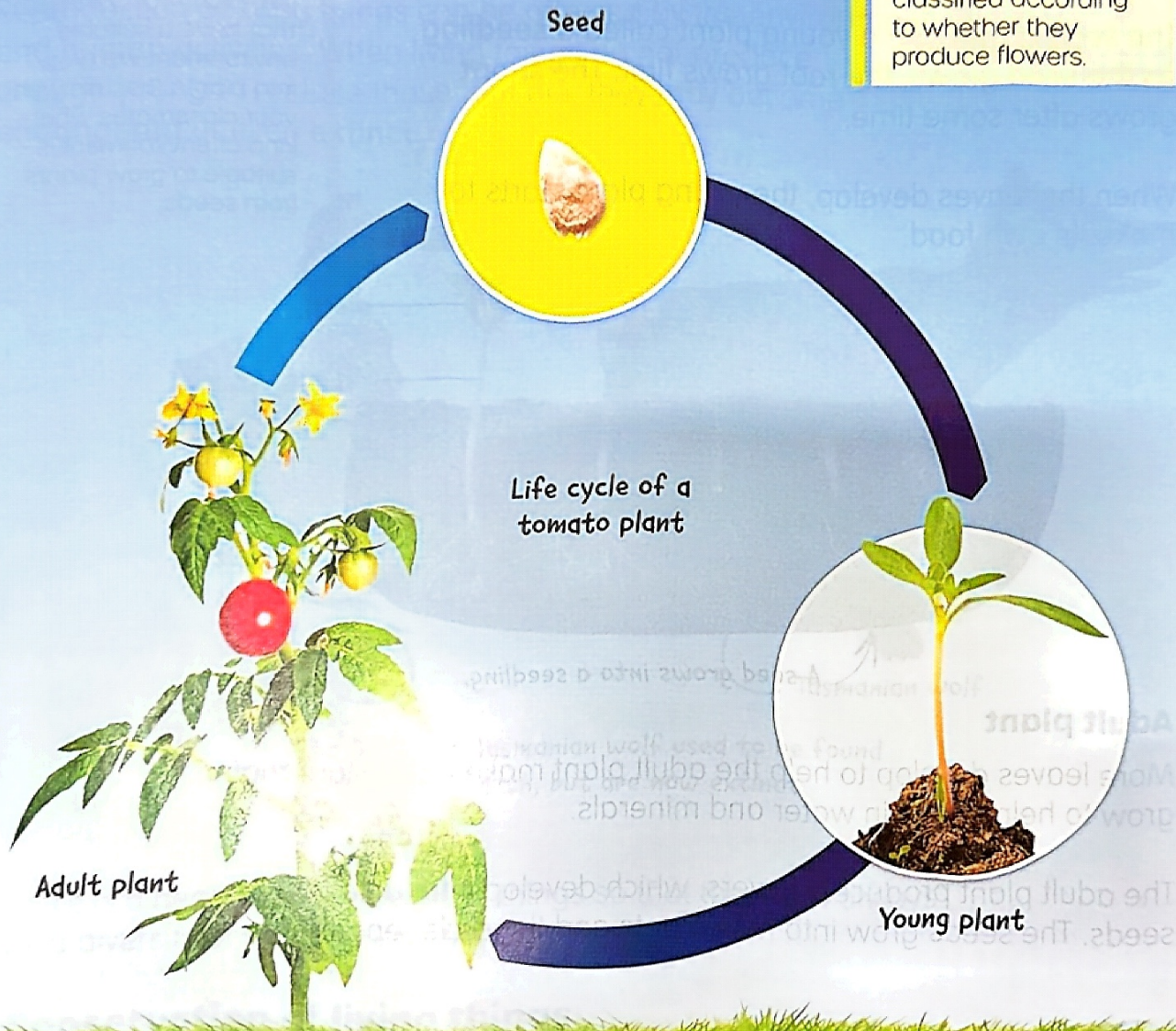
Life cycle of a flowering plant

Many flowering plants, such as the tomato plant, reproduce from seeds. They have three stages in their life cycles — seed, young plant and adult plant.



FLASHBACK

Recall from Textbook 3 Unit 4: Plants can be classified according to whether they produce flowers.



How are the life cycles of the different living things similar?
How are the life cycles different?



Seed

The seed needs air, water and warmth to start to grow or **germinate**.

Young plant

The seed grows into a young plant called a **seedling**. As the seed grows, the root grows first. The **shoot** grows after some time.

When the leaves develop, the young plant starts to make its own food.



EXPLORE

Not all seeds grow into plants. Only those that have a suitable environment will grow into plants. Discuss with your classmates what kind of environment is suitable to grow plants from seeds.



Adult plant

More leaves develop to help the adult plant make food. More roots grow to help it take in water and minerals.

The adult plant produces flowers, which develop into fruits with seeds. The seeds grow into new plants and the cycle repeats.



QUICK CHECK

1. Name the stages in the life cycle of a human.
2. What is the young of a frog called? State two differences between the young and the adult frog.
3. What are the stages in the life cycle of a tomato plant?

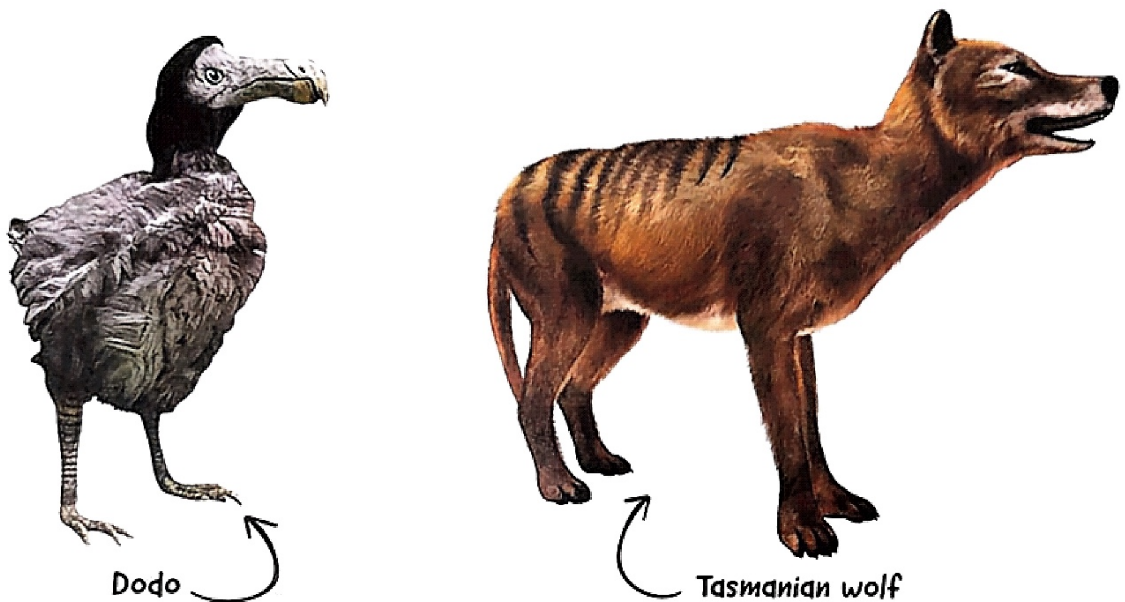


Activity Book

► Activity 3, pages 31-33

Why Is It Important to Study the Life Cycles of Living Things?

Reproduction of living things can be affected by the environment and human activities. When living things do not produce enough young to replace those that die, they may become **endangered** or even **extinct**.



The Dodo and Tasmanian wolf used to be found on the Earth, but are now extinct.

There is a need to **conserve** living things so that there will always be a diversity of living things on the Earth.

Conservation of living things

As living things grow and develop, they face different threats and their needs change. Some living things grow slowly and some produce a small number of young. By understanding the life cycles of living things, we can help conserve endangered animals such as sea turtles and orangutans.

Sea turtles

Sea turtles live in the sea. They are hunted for their beautiful shells and meat. Their eggs are eaten by humans and other animals. How can we help conserve sea turtles?

Only a few out of a thousand young sea turtles will survive to reproduce.

We can help protect sea turtles by keeping the place where they live clean. People should also be banned from hunting the sea turtles.

Female sea turtles lay eggs on the beach once in every two to three years. The eggs take about two months to hatch.

We should not touch or collect sea turtle eggs. This will help the eggs to hatch safely.



LANGUAGE CONNECT

In groups, write slogans to support the conservation of endangered animals. Share your slogans with the class.



Orangutans

Clearing of forests destroys the place where orangutans live. We can conserve orangutans by protecting forests.

Orangutans spend their lives in trees.

We can help protect forests by setting up national parks and protected areas in the forests. In this way, orangutans can continue to live.

Female orangutans usually give birth to one young at a time, once in every five to eight years. The young stays with the mother for a few years before it is able to survive on its own.

When forests are cleared, the young are separated from their mothers. Some countries set up protected places to care for the young of orangutans. The young are returned to forests when they are big and strong enough.



QUICK CHECK

1. Why is there a need to conserve living things?
2. State one way in which we can help conserve sea turtles.
3. State one way in which we can help conserve orangutans.

What We Have Learnt

Life Cycles

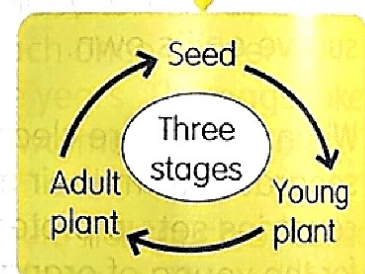
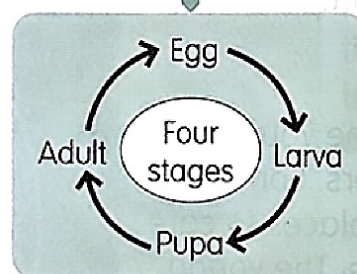
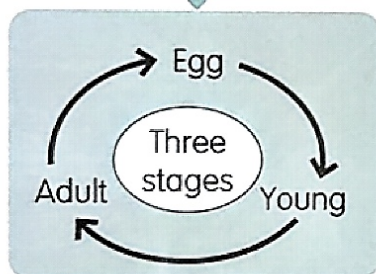
include

LIFE CYCLES OF ANIMALS

LIFE CYCLES OF FLOWERING PLANTS

which can be

which can be



for example

for example

for example

- Human
- Cat

Frog

- Butterfly
- Mosquito

Tomato plant

which

which

which

Reproduce by giving birth

Reproduces by laying eggs

Reproduce by laying eggs

Test Yourself

1. Look at the two groups of animals.



The animals have been classified based on the way they reproduce.
Give a suitable heading for each group.

2. Which two animals have four stages in their life cycles?

- (1) Cat and frog
- (2) Cat and butterfly
- (3) Butterfly and frog
- (4) Butterfly and mosquito

3. Joe kept a butterfly egg and observed it for a period of time. The table shows his observations.

Date	Observation
2 January	The egg was laid on a leaf.
7 January	The egg hatched.
17 January	The young started to hang upside down.
21 January	The young was fully wrapped in a hard case.
7 February	The hard case split open and a butterfly came out.

- (a) On which date did the young of the butterfly stop eating?
- (b) Draw a diagram to show the life cycle of a butterfly.
4. Sea turtles are endangered animals. Sometimes, scientists use special protected places called hatcheries to hatch sea turtle eggs. The young sea turtles are then returned to the sea.
- (a) How does hatching the eggs in hatcheries help conserve sea turtles?
- (b) State **one** way in which we can help protect sea turtles.



Activity Book

Let's Review, pages 34-37

SCIENCE GLOSSARY

Caterpillar

Young of a butterfly

Conserve

Save or protect

Develop

Grow and become bigger

Endangered

In danger of becoming extinct

Extinct

Cannot be found on the Earth anymore

Germinate

Start to grow

Larva

Stage in the life cycle of some insects, after the egg hatches

Life cycle

Cycle that shows the stages of growth and development of a living thing

Moult

Grow a new skin and shed the old one

Pupa

Stage between the larval and adult stages in the life cycles of some insects

Reproduce

Have young

Seedling

Young plant

Shoot

Part of a plant that includes the stem and leaves

Stage

Part in a process

Tadpole

Young of a frog

Wiggler

Larva of a mosquito

Science Today



Mosquitoes are pests. They can cause diseases such as malaria and dengue fever, which make us very sick.

Scientists made use of their understanding of the life cycle of mosquitoes and came up with the sterile insect technique (SIT) to control the number of mosquitoes.

A special device is used to release a large number of male mosquitoes that do not reproduce. As a result, the eggs that the female mosquitoes lay will not hatch. This breaks the life cycle of some mosquitoes and thus reduces the number of mosquitoes.



↖
The Aedes mosquito can spread dengue fever.